

NATURAL GAS PROCESSING

Design Basis, Design Basis Accident, and Military Action Extension

DBA-GAS-FC2: Natural Gas Processing Facility Class

Issued under DBA-GAS Sector Framework (v1.6) and DBA-EN-SF Energy Sector Framework (v0.7)

Version 0.2 — DRAFT

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For review by engineers, regulators, asset owners, and policy analysts responsible for protecting energy sector critical infrastructure.

Preface

The natural gas processing plant occupies the most analytically underappreciated position in the natural gas supply chain. It sits between the wellhead and the transmission system, visible to neither the pipeline engineer nor the end-user, performing the unglamorous but irreplaceable function of separating what the transmission system cannot accept — natural gas liquids, water, hydrogen sulfide, carbon dioxide — from what it can. When a major processing plant is destroyed, upstream production has nowhere to go: wellheads must be shut in, not because the wells are damaged, but because the gas they produce cannot be brought to pipeline quality. The disruption propagates upstream and downstream simultaneously, and it does so silently — without the visible fire of a refinery attack or the measurable pressure drop of a pipeline rupture.

Version 0.1 of this instrument established the DBA-MA scenarios for deliberate military action against natural gas processing infrastructure: the coordinated cyber-kinetic attack on a turbo-expander complex, the sour gas H₂S mass casualty scenario, and the Permian Basin processing corridor cluster attack. It also established the series-first recognition that the Bright Line in continuous-process optimization facilities must constrain the plant's own commercial optimization logic as well as adversarial manipulation — a non-adversarial structural risk that version 0.1 identified but did not develop into a formal Design Basis layer.

Version 0.2 completes the three-layer instrument by adding the non-adversarial DB and DBA layers that the MA extension requires as its foundation. The most analytically significant addition is the formal PPSD acceptance criterion framework: Pipeline Pressure Supply Duration was defined as a metric in v0.1 §1.3 and sanctioned at the DBA-EN-SF level in v0.6 §4.4, but its Zone 1/2/3 escalation structure — the acceptance criterion architecture that operators and regulators can act on — is established here for the first time. The PPSD framework is the FC2 analog to the SABD framework established in DBA-ES-CntI Co v0.2: a functional metric that bounds the consequence of the governing failure mode and structures the escalation response.

Section 1 — Introduction

1.1 Purpose

This document establishes the full three-layer Design Basis — Design Basis Accident — Military Action framework for the natural gas processing facility class. Version 0.1 established the DBA-MA scenarios directly; version 0.2 adds the non-adversarial DB and DBA layers that the MA extension requires as its foundation, completing the three-layer progression mandated by DBA-EN-SF §2.1 for all facility-class instruments.

The addition of the DB and DBA layers is not merely structural compliance with the series architecture. The non-adversarial bounding events for a processing plant — equipment failure, natural hazard, and operational upset — produce consequences that are analytically prior to the MA extension: the DBA-MA scenario derives its consequence envelope from the non-

adversarial DBA, and the MA characteristics (intentionality, persistence, precision, denial) are defined by what they add above and beyond the non-adversarial baseline. A processing plant that cannot recover from a non-adversarial cold box failure cannot be expected to recover faster from an intentional cold box destruction, and the PPSD acceptance criterion framework must bound both.

1.2 Scope and Applicability

This document addresses natural gas processing facilities: plants that receive raw wellhead gas and deliver pipeline-quality dry gas and NGL products through NGL extraction, acid gas removal, dehydration, and NGL fractionation operations. The facility class encompasses both gathering-system processing plants in producing fields and straddle plants on transmission systems. The primary reference environments are the Permian Basin, the Midcontinent, and the Gulf Coast processing corridors.

Natural gas transmission pipelines and compressor stations are addressed in DBA-GAS-FC1. Underground storage is addressed in DBA-GAS-FC3. Local distribution is addressed in DBA-GAS-FC4. LNG terminals are addressed in DBA-GAS-UC-LNG. The NGL consequence chain is within the scope of this document; downstream petrochemical facility consequences are noted as Tier 3 cross-sector cascades.

1.3 PPSD — Governing Consequence Metric

Pipeline Pressure Supply Duration (PPSD) is the governing consequence metric for the natural gas processing facility class. PPSD is defined as the duration of pipeline supply pressure consequence following a processing plant outage — the period during which the loss of processing capacity at a specific plant or cluster of plants produces a measurable reduction in pipeline delivery pressure or gas quality at the interconnect with the transmission system.

PPSD was established in this document at v0.1 §1.3 and sanctioned at the DBA-EN-SF level in v0.6 §4.4. Its sequential relationship to GSD (Generation Supply Duration) is formally established in DBA-EN-UC-1 §5.1: PPSD measures the upstream consequence of processing plant loss propagating through the transmission system; GSD measures the downstream consequence of that pressure reduction on gas-fired generation. These metrics are sequential, not additive — PPSD is the input to GSD, not a parallel metric.

Version 0.2 establishes the PPSD acceptance criterion framework in §4.3 with a Zone 1/2/3 escalation structure. This is the analytical completion that v0.1 deferred: PPSD as a defined metric is necessary but not sufficient for Design Basis purposes; PPSD with acceptance criteria and escalation thresholds is the governing instrument.

1.4 Relationship to Governing Documents

This document is issued under the DBA-GAS Sector Framework (v1.6) and DBA-EN-SF: Energy Sector Design Basis Framework (v0.7). The threat taxonomy, bounding methodology, consequence analysis structure, and adequacy standards are inherited from the sector framework. The PPSD → GSD sequential relationship is governed by DBA-EN-UC-1 §5.1. The Cross-Sector Consequence Elevation Rule (DBA-EN-SF §3.4) applies when the processing plant is the sole supply path for a Tier 2 or Tier 3 designated facility.

The Bright Line and Command Broker architecture is inherited from DBA-EN-SF and DBA-ES §3.4 and applied with processing-specific adaptation in §6. The non-adversarial optimization-safety boundary established first in this instrument at v0.1 §3.3 has been propagated to DBA-OIL-FC1 §3.3 (v1.3) and DBA-GAS-FC1 §8.5 (v1.6).

1.5 Position in Series

Document: DBA-GAS-FC2. Version: 0.2 — Full three-layer instrument.

Parent: DBA-GAS Sector Framework v1.6. Series Grandparent: DBA-EN-SF v0.7.

Gap closure: Completion of this instrument closes Gap 2.2 in the DBA-EN series gap register (DBA-GAS FC2/FC3/FC4 — DB Layer Only).

Section 2 — Processing Plant Profile and Asset Taxonomy (Retained from v0.1)

The processing plant profile and asset taxonomy established in v0.1 §2 are carried forward without substantive change. The governing text remains in v0.1 §§2.1 through 2.3. Where v0.1 and v0.2 conflict, v0.2 governs. A summary reference is provided below.

2.1 Asset Taxonomy Summary

Asset Class	Key Components	DB Bounding Concern	RT_HVLLC Applicability
Inlet separation and compression	Inlet separators, inlet compressors, scrubbers	Inlet compressor failure gates all downstream processing regardless of unit status.	Not primary — compressor lead time 6–12 months; not governing vs. cold box
NGL extraction — turbo-expander	Cold box (cryogenic heat exchanger), expander-compressor, expander wheel	Cold box destruction is the governing RT_HVLLC event: 8–16 months fabrication lead time. Series maximum for	RT_HVLLC: 8–16 months (cold box). Governing recovery constraint for Scenarios A and C

		gas processing	
Acid gas removal	Amine treating units (MEA/DEA/MDEA), thermal oxidizer, sulfur recovery unit	H ₂ S release from failed amine unit or thermal oxidizer — acute toxic hazard with off-site lethal vapor cloud potential	Not RT_HVLLC governed — amine systems replaceable in 4–8 weeks.
Dehydration	Glycol absorption units, molecular sieve units	Molecular sieve failure in the turbo-expander plant can force a reduction in the operating temperature limit.	Not individually RT_HVLLC governed.
NGL fractionation	Deethanizer, depropanizer, debutanizer columns; reboilers, condensers	Column section destruction — 4–8 month replacement lead time. NGL product delivery stops even if upstream extraction continues	RT_HVLLC: 4–8 months (column sections). Parallel path to cold box recovery
DCS and APC	Distributed control system, advanced process control (APC), safety instrumented system (SIS)	APC operating-band encroachment on SIS trip points — non-adversarial structural risk. DCS compromise as a cyber attack pathway	DCS reconstitution: 4–6 weeks. Governs Phase 1 recovery for all cyber-component scenarios

2.2 The Dual Consequence Chain

The defining characteristic of the processing facility class is the simultaneous dual consequence chain: gas supply curtailment propagating through the transmission system (PPSD pathway), and NGL supply curtailment propagating through the petrochemical feedstock system. These chains operate independently and simultaneously. The DB and DBA layers in §§ 3 and 4 address both chains; the MA extension in §5 addresses the adversary's ability to maximize compound disruption across both chains by target selection.

Section 3 — Design Basis: Non-Adversarial Bounding Events

3.1 DB Layer Purpose and Structure

The Design Basis layer establishes the cause-agnostic postulated events that a natural gas processing facility must demonstrate it can withstand. These events are defined without reference to adversarial intent — they bound the consequence envelope that equipment failures, natural hazards, and operational upsets can produce. The DBA-MA extension in §5 then asks what the four MA characteristics (intentionality, persistence, precision, denial) add above and beyond this non-adversarial baseline.

Three event categories organize the DB layer for the processing facility class: equipment failure (EF), natural hazard (NH), and operational event (OE). A fourth category, cross-sector cascade (XC), is inherited from the DBA-EN-SF and DBA-GAS sector parent and addresses processing plant consequences that originate in or propagate to adjacent facility classes.

3.2 Equipment Failure (EF) Postulated Events

Code	Event	Initiating Condition	Governing Concern
EF-1	Inlet compressor failure — single train	Mechanical failure of the primary inlet compressor train: bearing failure, seal failure, or driver loss. Gas flow to the processing plant drops to zero or is reduced to available bypass capacity.	Inlet compressor loss gates all downstream processing. If no bypass capacity is available, the plant shuts down. Upstream gathering system pressure rises toward MAOP; wellhead back-pressure increases. PPSD begins at the moment plant throughput drops below the transmission system delivery pressure floor.

EF-2	Cold box mechanical failure — turbo-expander plant	Cryogenic heat exchanger (cold box) mechanical failure from thermal stress, brazed aluminum plate-fin failure, or pressure excursion. The plant cryogenic section must be warmed up and taken out of service. NGL extraction ceases.	Cold box failure is the governing single-component RT_HVLLC event for the processing facility class: 8–16 month fabrication lead time for a plant-specific replacement unit. Partial NGL recovery may continue via JT expansion bypass, but ethane recovery drops from >90% to near zero. Gas quality to transmission is maintained; NGL production is curtailed.
EF-3	SIS spurious trip — process unit shutdown	Safety instrumented system actuates on a spurious signal, initiating emergency shutdown of a process unit. Depending on which unit is tripped, the plant may partially or fully shut down pending SIS investigation and restart authorization.	Spurious SIS trips are the most frequent cause of unplanned processing plant outages. Their DB relevance is the restart authorization requirement: a unit cannot be restarted after an SIS trip without investigation of the trip cause and confirmation that the SIS functioned as designed. The restart authorization time — not the trip itself — determines the PPSD contribution of a spurious trip event.
EF-4	DCS hardware failure — loss of process control	A distributed control system hardware failure removes	DCS hardware failure is the non-adversarial analog to

		automated process control from one or more processing units. Operators transition to manual control using local panel instrumentation. If DCS failure is complete, the plant may need to shut down pending DCS restoration.	the cyber attack scenario in the MA extension. The governing concern is the manual operations mode capability: can the plant maintain safe operations and some level of throughput on local panel control while DCS hardware is restored or replaced? Manual operations mode capability is a DB requirement, not a contingency.
EF-5	Amine unit failure — sour gas plant	Amine treating unit mechanical failure (pump failure, heat exchanger fouling, stripper column internals damage) at a sour gas processing plant. Acid gas removal capacity is reduced or eliminated. Gas cannot meet pipeline H₂S specification; plant must reduce throughput or shut in.	For sour gas plants, amine unit failure is a gas quality event as well as a throughput event: gas that does not meet pipeline H₂S specification cannot be delivered to the transmission system. If no bypass or alternative amine capacity is available, throughput shuts in regardless of other unit availability. H₂S specification failure is a gas quality PPSD trigger, not only a volume PPSD trigger.

3.3 Natural Hazard (NH) Postulated Events

Code	Event	Initiating Condition	Governing Concern
NH-1	Freeze event — hydrate formation	Ambient temperatures below	Hydrate events are the natural hazard

	and pipeline blockage	the hydrate formation temperature for the plant operating pressure, combined with loss or degradation of dehydration capacity, produce hydrate plugs in the cryogenic section or in gathering system piping. Plant throughput is reduced or eliminated until hydrate plugs are removed.	analog to the Winter Storm Uri scenario: low ambient temperature combined with reduced dehydration capability or loss of heat tracing produces hydrate plugs that block process flow. Hydrate removal by depressurization and warming is a time-consuming process, hours to days, depending on plug location and severity. Gathering system hydrate plugs upstream of the plant compound the processing plant PSD with gathering system delays.
NH-2	Flooding — electrical infrastructure submersion	Flood event submerges the plant electrical switchgear, MCC rooms, or substation, removing power from process units and DCS. The plant shuts down. Restart requires electrical system inspection, testing, and potential equipment replacement before energization.	Processing plants in Gulf Coast and Midcontinent locations are subject to flooding from tropical weather events, river flooding, and flash flooding. Electrical infrastructure submersion is a more consequential event than process unit flooding because it removes the ability to restart any process unit until electrical systems are restored and tested. The electrical inspection and testing

			requirement — not physical flooding damage — typically governs the restart timeline.
NH-3	Seismic event — piping and vessel integrity	Seismic ground motion produces stress on cryogenic piping, pressure vessels, and column foundations. Damage may range from instrument connections and small-bore piping to structural damage to column foundations or cold box support structures.	Processing plants in seismically active areas — portions of the Midcontinent, particularly Oklahoma, and the Permian Basin fringe — are subject to induced seismicity from wastewater disposal. A seismic event that produces even minor damage to cryogenic piping requires inspection of all affected systems before restart: the combination of cryogenic temperatures and hydrocarbon service makes undetected small leaks a fire and explosion hazard.
NH-4	Extreme heat — cooling water and air cooler limitation	Sustained extreme ambient temperatures reduce air cooler capacity below the cooling duty required to maintain NGL fractionation column condenser performance. NGL product purity degrades, or throughput must be reduced to maintain specification.	Processing plants in the Southwest and Gulf Coast operate air-cooled condensers on fractionation columns. Extreme heat events — ambient temperatures above the air cooler design condition — reduce condenser capacity and can force throughput reductions to maintain NGL purity

			specifications. Extended extreme heat events can produce multi-day throughput reductions with PPSD consequences if NGL storage capacity is limited.
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3.4 Operational Event (OE) Postulated Events

Code	Event	Initiating Condition	Governing Concern
OE-1	Feed gas composition excursion — rich gas upset.	Raw wellhead gas composition shifts outside the plant design envelope: higher-than-design NGL content (rich gas excursion), higher-than-design H ₂ S or CO ₂ content, or presence of contaminants (mercury, BTEX, heavy hydrocarbons) that are not managed by installed pretreatment. The plant must reduce throughput or shut down to protect downstream units.	Feed gas composition excursions are the most operationally common cause of unplanned processing plant constraints. Producing wells change composition over their life cycle; seasonal variations affect raw gas composition in some formations. The DB concern is the plant's demonstrated ability to detect composition changes before they exceed unit operating limits — not the composition change itself, which is a normal operational condition.
OE-2	APC operating-band encroachment on SIS trip points	The advanced process control system operating band, through incremental optimization adjustments, progressively approaches SIS trip points without triggering the PSM MOC process. A normal	This is the series-first non-adversarial structural risk identified in v0.1 §3.3 and established as the governing non-adversarial DB event for the processing

		process perturbation that would have been absorbed with an adequate margin triggers an SIS trip under the reduced-margin APC configuration.	facility class. It is placed in the OE category rather than EF because it originates in operational decision-making — the incremental APC optimization adjustments — rather than in equipment failure. The Bright Line must constrain the APC optimization logic as a DB requirement, not only as an MA response. This event is what the OSHA PSM MOC gap identified in v0.1 §7.2 allows to occur without a formal safety review.
OE-3	NGL product specification failure — fractionation upset	Fractionation column control upset produces off-specification NGL product: ethane content in propane product above specification, propane content in ethane above specification, or water or contaminants in any NGL product stream. The product cannot be delivered to the pipeline or storage; it must be recycled or flared.	NGL product specification failure is a revenue and operational event at low severity, but a PPSD event if it forces the fractionation column shutdown for corrective action. The governing DB concern is the fractionation column's control system stability under feed rate changes and feed composition variation — the same operating conditions that the MA extension targets as

			the precision attack vector.
OE-4	Gathering system supply interruption — upstream curtailment	Upstream gathering system operator curtails gas deliveries to the processing plant due to well integrity events, gathering line failures, or severe weather affecting field production. Plant throughput drops below the minimum stable operating rate for cryogenic units; partial or full shutdown is required.	Gathering system supply interruptions are the upstream analog to the transmission system supply disruption that PPSD measures downstream. The DB concern is the plant's minimum stable operating rate: below a certain feed rate, cryogenic units cannot maintain operating temperature. They must be warmed up — a process that takes hours and adds to the PPSD clock even after upstream supply resumes.

3.5 Cross-Sector Cascade (XC) Postulated Events

Code	Event	Source	FC2 Consequence
XC-1	Electric supply loss to processing plant — electric-drive equipment	The electric sector (DBA-ES) event is producing a supply loss to the processing plant. Electric-drive inlet compressors, refrigeration compressors, and DCS power supply affected.	The processing plant shuts down on loss of electric supply. Gas-drive plants maintain throughput. The XC-1 event produces a PPSD consequence from a non-gas-sector initiating condition — the gas-electric bidirectional cascade pathway running in the electricity-to-gas direction.

XC-2	Transmission system back-pressure event — downstream constraint	The DBA-GAS-FC1 event (transmission compressor station outage) produces elevated back-pressure on the processing plant delivery point. The plant must reduce throughput or curtail operations to maintain delivery pressure within transmission system operating limits.	XC-2 is the downstream analog to XC-1: a transmission system event produces a processing plant constraint. The plant cannot deliver gas at the required pressure; it must curtail throughput. PPSD from XC-2 is a cascade consequence of the FC1 event, not an independent FC2 initiating condition.
XC-3	Petroleum sector NGL pipeline outage — NGL storage overflow	DBA-OIL-FC4 event (NGL pipeline outage) prevents NGL product delivery from the processing plant. NGL storage fills; fractionation must curtail, or the plant must recycle Y-grade NGL into the gas stream.	XC-3 bridges the gas and petroleum sectors through the NGL pipeline system. An NGL pipeline outage that fills processing plant NGL storage forces the plant to curtail NGL production, which may force a reduction in gas throughput if NGL production is required for gas quality reasons.

Section 4 — Design Basis Accident: Non-Adversarial Consequence Envelope

4.1 Bounding Event Identification

The DBA layer identifies the bounding events — the most severe non-adversarial postulated events from the DB layer — that define the consequence envelope within which protective measures and recovery architecture must perform. For the processing facility class, three bounding events govern.

Bounding Event	Source Event	Governing Consequence	Metric
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Cold box mechanical failure — complete NGL extraction loss	EF-2	Loss of ethane and NGL extraction. Gas quality to transmission is maintained. NGL product delivery ceases for RT_HVLLC interval (8–16 months for cold box replacement). Downstream ethylene cracker and NGL market disruption.	RT_HVLLC: 8–16 months
APC operating-band SIS trip — full plant shutdown	OE-2	Plant shuts down on SIS actuation. PPSD begins. Gas supply to the downstream transmission system is curtailed. NGL production ceases. Recovery requires SIS investigation, authorization, and restart — hours to days depending on trip complexity.	PPSD: hours to days (SIS restart authorization governs)
Processing corridor cluster outage — multi-plant simultaneous	NH-1 / EF-1 compound, or OE-4 cascade	Simultaneous outage of multiple processing plants in a producing corridor exceeds regional adaptive capacity. PPSD propagates into GSD as gas-fired generation curtails. Associated gas flaring or oil production curtailment in the Permian Basin paradigm.	PPSD (multi-plant): days to weeks. GSD (downstream): triggered when PPSD exceeds transmission system buffer capacity

4.2 Consequence Taxonomy

The three-tier consequence taxonomy from DBA-EN-SF §2.3 applies to the processing facility class as follows.

Tier	Processing Facility Consequence Definition	Bounding Event	Governing Metric
Tier 1	Facility-level consequence: single processing plant throughput reduction or shutdown. Gas supply to the transmission system has been reduced but not eliminated. The downstream transmission system absorbs the reduction within normal operating flexibility. NGL production at the affected plant has been curtailed.	EF-1 (single inlet compressor), EF-3 (spurious SIS trip), OE-1 (feed gas composition excursion)	PPSD Zone 1: plant-level throughput reduction, transmission system not materially affected
Tier 2	Grid-level consequence: processing plant outage produces measurable reduction in pipeline delivery pressure or gas quality at the transmission interconnect. The downstream transmission system cannot fully absorb the reduction without flow rerouting or supply allocation. Gas-fired generation at risk of curtailment if PSD extends beyond buffer	EF-2 (cold box failure), EF-5 (amine unit failure — sour plant), NH-1 (hydrate event), OE-2 (APC SIS trip)	PPSD Zone 2: transmission system pressure impact confirmed. GSD risk is elevated if PSD persists beyond T ₁ .

	capacity.		
Tier 3	Cross-sector cascading consequence: processing plant outage triggers GSD at one or more compressor stations; gas-fired generation curtails; electric sector N-1 contingency management is degraded. NGL supply curtailment produces a petrochemical feedstock cascade. Permian Basin corridor attack forces oil production curtailment.	Multi-plant cluster outage (Permian Basin paradigm), XC-1 compound (electric supply loss to processing plant during high-gas-generation period)	PPSD Zone 3: GSD triggered. Cross-sector consequence elevation rule (DBA-EN-SF §3.4) applied —Federal Coordination Threshold notification.

4.3 PPSD Acceptance Criterion Framework

The PPSD acceptance criterion framework establishes the Zone 1/2/3 escalation structure for the Pipeline Pressure Supply Duration metric. PPSD begins at the moment a processing plant event produces a measurable reduction in pipeline delivery pressure or gas quality below the transmission system's minimum acceptance criteria at the plant's delivery point. PPSD ends at the moment pipeline delivery pressure and gas quality return to within transmission system specifications and are confirmed stable.

PPSD Zone	Condition	Required Response	Escalation Trigger
Zone 1 (0 to T ₁)	Plant throughput reduced, but transmission system delivery pressure is within the normal operating range. Gas quality specifications met. PPSD clock is running, but no downstream system	Plant operator: initiate restart sequence or unit isolation. Gather system operator: notified, monitoring upstream pressure. Transmission operator: monitoring delivery pressure, no	Confirmed delivery pressure below transmission minimum, or gas quality specification failure at delivery point → Zone 2.

	impact has been confirmed.	action required.	
Zone 2 (T ₁ to T ₂)	The transmission system delivery pressure was impacted. Downstream flows require rerouting or supply allocation adjustments. Gas-fired generation at connected compressor stations is operating on a reduced supply margin. NGL product delivery curtailed.	PHMSA and CISA notification (if cyber component suspected). Transmission operator activates supply rerouting procedures. GSD risk assessment initiated at downstream compressor stations. Emergency NGL supply allocation coordination initiated.	GSD confirmed at one or more downstream compressor stations, or a multi-plant cluster event confirmed → Zone 3.
Zone 3 (beyond T ₂)	GSD triggered at the downstream compressor stations. Gas-fired generation curtailing. Electric sector N-1 contingency management at risk. The NGL petrochemical feedstock cascade is being developed. Permian oil production curtailment risk if a Permian cluster event occurs.	Federal Coordination Threshold notification. DOE ESF-12 activation. Cross-sector consequence elevation rule evaluated (DBA-EN-SF §3.4). Emergency flaring authorization coordination (Permian paradigm). Mutual aid and alternative supply coordination across all affected sectors.	PPSD Zone 3 conditions are the Tier 3 consequence trigger. No further escalation zone — Zone 3 is the maximum consequence condition for the processing facility class.

T₁ and T₂ calibration: T₁ is the transmission system buffer duration — the period for which the downstream transmission system can maintain delivery pressure within operating limits without additional supply. T₁ is facility-specific and depends on pipeline inventory, downstream compressor station operating flexibility, and seasonal demand conditions. Series-level guidance: T₁ is typically 4 to 24 hours for a single-plant outage; shorter for cluster events. T₂ is the GSD trigger threshold — the PSD duration at which downstream compressor stations begin curtailing gas-fired generation. T₂ is governed by DBA-GAS-FC1 §5 criteria and the compressor station-specific minimum inlet pressure floor.

4.4 RT_HVLLC Application

The RT_HVLLC metric applies to the cold box (cryogenic heat exchanger) as the governing high-value long-lead component for the turbo-expander processing facility class. RT_HVLLC for the cold box is 8 to 16 months, reflecting the engineering and fabrication time for a plant-specific cryogenic heat exchanger by one of the small number of global manufacturers capable of producing large-scale processing plant cold boxes.

RT_HVLLC for the processing facility class is not the same as the PPSD Zone 3 duration. PPSD measures the supply pressure consequence at the transmission interconnect, which begins immediately on plant outage and may extend for the full RT_HVLLC period if no bypass capacity is available. RT_HVLLC measures the time to restore the facility's full NGL extraction capability. These are sequential: PPSD Zone 3 may resolve before RT_HVLLC expires if partial plant restart (JT bypass, undamaged train capacity) restores gas quality and volume to transmission system specifications. At the same time, the cold box is still being fabricated.

Section 5 — DBA-MA Military Action Extension

5.1 MA Extension Foundation

The DBA-MA extension applies the four MA characteristics — intentionality, persistence, precision, and denial — to the non-adversarial bounding events established in §§ 3 and 4. This is the same analytical structure as v0.1, now formally grounded in the DB and DBA layers. The three scenarios from v0.1 are carried forward with reference updates and a clarification of their relationship to the PPSD acceptance criterion framework.

The governing insight from v0.1 remains unchanged: the adversary who combines DCS pre-positioning with precision kinetic action against specific process units can simultaneously maximize both consequence chains — gas supply disruption (PPSD pathway) and NGL supply disruption (petrochemical feedstock pathway) — in a way that no non-adversarial event achieves. The MA characteristics add above the DBA baseline by enabling target selection, timing control, and denial of the recovery indicators that operators rely on.

5.2 Four MA Characteristics Applied to Processing Facility Class

MA Characteristic	Application	Governing Concern
Intentionality	Adversary selects processing plants whose simultaneous outage maximizes compound disruption across both consequence chains. Target selection uses public-domain data: FERC filings, EPA	Target selection intelligence is available without insider access. The Permian Basin concentration of processing capacity — where four or five major plants account for a disproportionate share of

	RMP submissions, pipeline flow data. Adversary identifies the minimum set of plants whose simultaneous outage exceeds regional adaptive capacity.	corridor capacity — makes the minimum target set small and the required intelligence public.
Persistence	Adversary maintains DCS compromise for a period sufficient to cause the APC operating band to encroach on SIS trip points before initiating the kinetic phase. Persistence also applies post-attack: adversary interdicts procurement of cold box replacement units or specialty fractionation column internals to extend RT_HVLLC.	Persistence at the DCS level extends the attack preparation timeline beyond the window that typical security monitoring covers. Persistence at the supply chain level — interdicting cold box fabrication or delivery — is the MA extension of the RT_HVLLC strategic reserve gap (v0.1 §7.3 / DBA-EN gap 4.3, now closed by RT_HVLLC Strategic Reserve WP v0.1).
Precision	Adversary targets the specific process units whose loss maximizes both PPSD and NGL curtailment duration: the cold box for NGL extraction loss; the fractionation deethanizer for ethane supply curtailment; the amine unit for gas quality failure at sour gas plants. Precision targeting avoids units that can be bypassed or replaced quickly.	The adversary's precision advantage over non-adversarial events is the selection of targets that simultaneously maximize both PPSD Zone 3 duration and RT_HVLLC. A random equipment failure is equally likely to hit a replaceable unit as a long-lead unit; a precision attack targets long-lead units exclusively.
Denial	MA-1 class attack: DCS pre-positioning causes APC operating band encroachment that appears to operators as a series of independent optimization adjustments, not as adversarial preparation. When the kinetic phase initiates, operators are denied the accurate system state picture needed to diagnose the compound event.	Denial in the processing plant context operates at two levels: denial of accurate process state (DCS manipulation before and during the attack), and denial of recovery capacity (supply chain interdiction of long-lead replacement components). Both forms of

		denial extend the PPSD Zone 3 duration beyond the non-adversarial baseline.
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5.3 Scenario A — Coordinated Cyber-Kinetic Attack on Turbo-Expander Complex (Retained from v0.1)

The governing DBA-MA scenario for the processing facility class. Adversary pre-positions in the DCS of a major turbo-expander processing complex; manipulates APC operating band to reduce margin to cold box temperature limits; initiates kinetic strike on the cold box and inlet compression when the APC manipulation has positioned the plant at reduced safety margin. The cold box is destroyed; the turbo-expander is damaged.

PPSD consequence: Zone 1 begins on plant shutdown. Zone 2 within T₁ as transmission delivery pressure drops. Zone 3 if GSD triggers at downstream compressor stations.

RT_HVLLC: 8–16 months for cold box replacement. NGL extraction loss: ethane recovery drops from >90% to near zero via JT bypass for the duration of cold box fabrication.

The DCS pre-positioning phase of Scenario A is now formally grounded in OE-2 (APC operating-band encroachment on SIS trip points) from the DB layer: the adversary exploits the same non-adversarial structural risk that OE-2 identifies. The difference between OE-2 and Scenario A is intentionality — OE-2 occurs through incremental commercial optimization decisions; Scenario A occurs through deliberate adversarial manipulation of the same APC architecture.

5.4 Scenario B — Sour Gas Processing Plant Attack — H₂S Mass Casualty Scenario (Retained from v0.1)

An attack targeting an amine treating unit or thermal oxidizer at a sour gas processing plant to produce an H₂S release with off-site lethal consequences. The H₂S release consequence envelope — lethal concentration zones, community evacuation requirements, emergency response resource demands — is established in the Tier 3 consequence taxonomy for this scenario.

PPSD consequence: Zone 1 begins on plant shutdown. Zone 2 if the affected plant is a significant supply source for the downstream transmission segment. The H₂S consequence does not extend the PPSD Zone duration but compounds the recovery complexity: the plant cannot restart until H₂S consequence management is complete, the release source is identified and isolated, and regulatory inspection authorizes restart.

5.5 Scenario C — Permian Basin Processing Corridor Cluster Attack (Retained from v0.1)

Simultaneous attacks on three to four major Permian Basin processing plants remove 600 MMcf/d to over 2 Bcf/d of processing capacity. Upstream wellhead shut-ins cascade across multiple gathering systems. Associated gas production curtailment forces oil production reductions of 500,000 to 1,500,000 barrels per day — the most significant domestic crude oil production reduction scenario in the DBA-OIL and DBA-GAS series.

PPSD consequence: Zone 3 immediately for a Scenario C event — the simultaneous multi-plant nature of the attack exceeds regional adaptive capacity. GSD triggered at downstream compressor stations serving Southwest power markets. Federal Coordination Threshold notification is a Scenario C immediate response requirement. The PPSD → GSD sequential relationship (DBA-EN-UC-1 §5.1) governs: Scenario C, PPSD is the initiating input to the downstream GSD cascade.

Section 6 — Dual Design Basis Architecture and AI Governance

6.1 AI Governance Interaction Points (Updated from v0.1)

The four AI governance interaction points established in v0.1 are retained. Version 0.2 adds a fifth interaction point derived from the DB layer: the formal grounding of the non-adversarial APC optimization-safety boundary (OE-2) as a Bright Line requirement applicable independent of adversarial intent.

Interaction Point	Bright Line Position	Requirement
APC optimization logic — non-adversarial boundary (NEW in v0.2, grounded in OE-2)	Below Bright Line — but Bright Line must be set at the margin to the SIS trip point, not at the SIS trip point itself.	The APC pre-authorized envelope must be defined with an explicit safety margin setback from all SIS trip points. APC logic may not autonomously adjust the pre-authorized envelope toward an SIS trip point without a human-authorized MOC process. This is a DB requirement independent of adversarial threat. This construction is the natural-gas-processing sector instantiation of FD-BL-D1 R3: a change to the pre-authorized envelope itself — including any movement of the operating region toward a protective setpoint (an SIS trip point) — is categorically excluded from advance discharge and routes instead to human-authorized MOC (in-the-moment authorization) or to offline requalification producing a new verified envelope. The setback is what keeps optimized operation inside a valid pre-authorized envelope; the prohibition on autonomous envelope movement toward a trip point is the R3 exclusion of envelope change from advance discharge.
APC optimization logic —	Below the Bright Line for	APC systems must include anomaly detection for systematic

adversarial manipulation boundary	deterministic APC; adversarial manipulation detection above the Bright Line	operating-band encroachment inconsistent with normal optimization behavior. Detected systematic encroachment requires human review before further APC adjustment is authorized. Per FD-BL-D1, this encroachment detection is an R2 input: R2 requires the acceptance set to be proven complete and sound over trajectory and rate — not only instantaneous state — so that no admissible sequence of individually acceptable optimization adjustments can compose over time into an unsafe approach to the trip points. Systematic-encroachment detection is the anti-ratcheting mechanism by which that trajectory/rate bound is enforced against incremental ratcheting. It is, equally, this sector's instantiation of FD-BL-D2 I3: the divergence between the measured trajectory and normal optimization behavior is the independent corroboration path by which the trustworthiness of the state feeding APC optimization is checked, so that a drifted or manipulated synthesis cannot serve as the sole basis for continued autonomous adjustment (see FD-BL-D2, indication integrity).
DCS process control — SIS interface	Below Bright Line (deterministic, formally testable)	SIS must be architecturally independent of APC and DCS optimization layers. SIS trip setpoints may not be modified by APC logic. This requirement is unchanged from v0.1.
Gen AI operational decision support	Above Bright Line — Command Broker required	Gen AI tools used in processing plant operational decision support (feed gas composition analysis, optimization recommendations, emergency procedure guidance) require a licensed process operator as Command Broker. Non-

		licensed personnel may not serve as Command Broker.
ML-assisted process optimization — model provenance	Below Bright Line is deterministic and auditable	ML models used in APC must have documented training data provenance, version history, and update authorization records. An ML model whose training data includes adversarially manipulated process historian data cannot be treated as a below-Bright-Line autonomous system until the data contamination is resolved.

6.2 The Optimization-Safety Boundary as Dual-DB Governing Principle

The OE-2 non-adversarial DB event and the Scenario A adversarial DCS manipulation scenario share the same underlying attack surface: the interface between the APC operating band and the SIS trip setpoints. This convergence — established first in this instrument at v0.1 §3.3 and now formally grounded in the DB layer at OE-2 — has been propagated to DBA-OIL-FC1 §3.3 (v1.3) and DBA-GAS-FC1 §8.5 (v1.6) as a series-wide analytical contribution.

The Dual Design Basis implication is that the AI Governance Design Basis for a processing plant must address the APC-SIS interface as a governance requirement independent of adversarial threat, because the same optimization pressure that creates OE-2 risk creates the attack surface that Scenario A exploits. A processing plant that has architecturally separated its APC optimization layer from its SIS layer has simultaneously addressed the non-adversarial OE-2 risk and reduced the adversarial Scenario A attack surface. This is the most cost-effective single remediation in the processing facility class design basis.

Section 7 — Protective Action Thresholds (Retained from v0.1)

The protective action thresholds, threat indicator taxonomy, decision authority framework, and federal coordination thresholds established in v0.1 §5 are retained without substantive change. The federal coordination thresholds are updated to reference the PPSD Zone 3 trigger as the governing federal notification criterion, consistent with the DBA-EN-SF §3.4 Cross-Sector Consequence Elevation Rule.

7.1 Federal Coordination Thresholds — PPSD Integration

Threshold	Trigger Condition	Notification Requirement
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PPSD Zone 3 — GSD confirmed	GSD confirmed at one or more downstream compressor stations as a consequence of processing plant PPSD	DOE ESF-12, PHMSA, CISA, FERC. Cross-sector consequence elevation rule (DBA-EN-SF §3.4) applied.
Kinetic event at processing facility	Confirmed kinetic strike at any major natural gas processing facility	FBI, DOD (if state-actor attribution suspected), CISA, PHMSA, DOE ESF-12.
DCS compromise — nation-state attribution	Confirmed DCS compromise with suspected nation-state attribution	CISA, FBI, DOD. Parallel notification to sector peer processing operators through ISAC.
H2S off-site release	H2S release affecting or threatening off-site areas above threshold concentration	EPA, local emergency management, and state emergency management—community notification per emergency response plan.
Cluster event — multi-plant	Simultaneous security incidents at two or more processing plants within a producing corridor within 72 hours	All of the above plus: Texas Railroad Commission (Permian events), DOE ESF-12 supply emergency declaration evaluation.

Section 8 — Recovery Architecture (Retained from v0.1, Updated for v0.2)

The recovery architecture established in v0.1 §6 is retained. Version 0.2 adds explicit references to the PPSD Zone transitions as recovery milestones, aligning the recovery sequencing with the acceptance criterion framework in §4.3.

8.1 Recovery Phase — PPSD Zone Alignment

Recovery Phase	Timeline	PPSD Zone Target	Governing Constraint
Phase 1 — Immediate	0–72 hours	Exit Zone 3, achieve Zone 2 if a partial restart is possible	Emergency shutdown and stabilization. DCS forensic investigation initiated. Cold box damage assessment. Procurement inquiry for cold box initiated

			(Phase 1 — governs critical path). Upstream gathering system and downstream transmission operator notification. PPSD Zone 3 federal notifications issued.
Phase 2 — Short-term	72 hours to 8 weeks	Sustain Zone 2, work toward Zone 1	DCS forensic investigation continuing. Cold box procurement underway. Partial plant restart on JT bypass or undamaged trains where available. NGL alternative supply coordination. Ethylene cracker feedstock contingency plans are being activated.
Phase 3 — Medium-term	8 weeks to RT_HVLLC expiry	Zone 1 on partial capacity; full capacity at RT_HVLLC	Cold box fabrication and delivery (8–16 months). Fractionation column replacement in parallel. DCS reconstitution and APC-SIS re-validation before NGL optimization resumes. Sequential restart and commissioning. PPSD clock stops when transmission delivery pressure and gas quality return to specification.

8.2 APC-SIS Re-Validation as Recovery Prerequisite

A processing plant that has experienced a DCS compromise — whether as part of Scenario A or as a standalone OE-2 event — must re-validate the APC operating band against SIS trip points before resuming optimized NGL recovery operations. This re-validation is a DB requirement established at §3.4 OE-2 and is a recovery prerequisite independent of the physical repair status of process units. A plant whose physical units are repaired but whose APC-SIS margin has not been re-validated is not permitted to resume APC-driven optimization under this design basis.

Section 9 — Gaps and Recommendations (Retained from v0.1, Updated for v0.2)

The five regulatory gap recommendations from v0.1 §7 are retained. Gap 4.3 (RT_HVLLC Strategic Reserve) has been closed at the series level by the RT_HVLLC Strategic Reserve White Paper v0.1, which addresses the cryogenic cold box procurement constraint. The remaining gaps are carried forward.

Gap	Description	Status	Lead Action
G-1	TSA pipeline cybersecurity directives do not cover independent processing plant DCS systems not owned by pipeline operators. Processing plant DCS has the same IT-OT boundary vulnerabilities as pipeline SCADA without the same mandatory cybersecurity framework.	Open	TSA, in coordination with CISA, should evaluate extending the pipeline cybersecurity directive framework to cover processing plant DCS at facilities whose outage would produce PPSD Zone 2 or Zone 3 consequences.
G-2	OSHA PSM does not explicitly require that APC pre-authorized envelope modifications reducing margin to SIS trip points go through the PSM MOC process. Incremental APC optimization adjustments can progressively reduce the safety margin without a formal safety review.	Open	OSHA should issue guidance clarifying that APC pre-authorized envelope modifications reducing the margin to any SIS trip point constitute a process change subject to PSM MOC requirements, regardless of the magnitude of individual adjustments. This is the regulatory gap that enables OE-2.

G-3	No strategic reserve exists for processing plant cryogenic cold box modules. Cold box fabrication lead time (8–16 months) is the governing RT_HVLLC constraint for the series' most consequential single-plant attack scenario.	CLOSED — addressed by RT_HVLLC Strategic Reserve WP v0.1	RT_HVLLC Strategic Reserve WP v0.1 establishes the policy case and program model. DOE/PHMSA engagement is required for implementation.
G-4	No emergency flaring authorization framework exists for accelerated permitting at the scale required by a Permian Basin multi-plant processing outage. Texas Railroad Commission and EPA flaring processes are not designed for emergency acceleration.	Open	Texas Railroad Commission and EPA should develop an emergency flaring authorization protocol triggered by a declared energy supply emergency, with defined sunset conditions tied to processing capacity restoration milestones.
G-5	No federal emergency response plan addresses the NGL and petrochemical feedstock consequences of a major processing plant attack. DOE ESF-12 has historically focused on electric power and natural gas transmission.	Open	DOE, in coordination with the American Chemistry Council and the National Propane Gas Association, should develop an NGL supply emergency response annex to ESF-12 covering consequence thresholds, federal coordination mechanisms, and alternative supply options for each major NGL product

			stream.
0.2 (housekeeping)	June 2026	T. Roxey	Housekeeping correction. DBA-EN-SF grandparent reference updated from v0.6 to v0.7 in cover lineage line, §1.5 issued-under sentence, and §1.5 parent/grandparent line. Note: the reference to “sanctioned at the DBA-EN-SF level in v0.6 §4.4” is retained as accurate historical record — PPSD was indeed sanctioned in v0.6; the current version reference is updated separately. No substantive analytical changes.

Section 10 — References

Regulatory and Standards References

U.S. Occupational Safety and Health Administration. Process Safety Management of Highly Hazardous Chemicals. 29 CFR 1910.119. Washington, D.C.: OSHA. Current version.

U.S. Environmental Protection Agency. Risk Management Program. 40 CFR Part 68. Washington, D.C.: EPA. Current version.

U.S. Department of Transportation, Pipeline and Hazardous Materials Safety Administration. Pipeline Safety Regulations. 49 CFR Parts 190–199. Washington, D.C.: PHMSA. Current version.

Transportation Security Administration. Pipeline Security Directives. Washington, D.C.: TSA. Current versions (SD-02A, SD-02B, SD-02C).

International Society of Automation / International Electrotechnical Commission. ISA/IEC 62443 Series: Security for Industrial Automation and Control Systems. Research Triangle Park: ISA. Current editions.

International Electrotechnical Commission. IEC 61511: Functional Safety — Safety Instrumented Systems for the Process Industry Sector. Geneva: IEC. Current edition.

Government and Policy References

Presidential Policy Directive 21 (PPD-21): Critical Infrastructure Security and Resilience. Washington, D.C.: The White House, February 2013.

U.S. Department of Energy. "The Department of Energy's Role in the Natural Gas Supply Chain." Washington, D.C.: DOE, 2021.

Federal Energy Regulatory Commission. "Report on Cyber Intrusion into Energy Sector Industrial Control Systems." Washington, D.C.: FERC, 2022.

Technical References

Campbell, John M. Gas Conditioning and Processing. 9th ed. Norman, Oklahoma: Campbell Petroleum Series, 2014. (Governing reference for NGL extraction, acid gas removal, and fractionation operations.)

Kidnay, Arthur J., William R. Parrish, and Daniel G. McCartney. Fundamentals of Natural Gas Processing. 3rd ed. Boca Raton: CRC Press, 2019.

Lees, Frank P. Loss Prevention in the Process Industries: Hazard Identification, Assessment and Control. 4th ed. Oxford: Butterworth-Heinemann, 2012. (H₂S consequence analysis and process hazard assessment.)

Series Parent and Cross-Reference Documents

Roxey, Timothy E. DBA-EN-SF: Energy Sector Design Basis Framework. Version 0.6 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Timothy E. DBA-GAS: Natural Gas Sector Framework. Version 1.6 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Timothy E. DBA-GAS-FC1: Natural Gas Transmission. Version 1.6 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Timothy E. DBA-EN-UC-1: Gas-Electric Cascade Consequence Model. Version 0.1 — DRAFT. Eclectic Technologies, June 2026. (PPSD-to-GSD sequential relationship, §5.1.)

Roxey, Timothy E. RT_HVLLC Strategic Reserve White Paper. Version 0.1 — DRAFT. Eclectic Technologies, June 2026. (Cryogenic cold box strategic reserve gap — G-3 closed.)

Roxey, Timothy E. DBA-OIL-FC1: Petroleum Refining. Version 1.3 — DRAFT. Eclectic Technologies, June 2026. (APC optimization-safety boundary propagated from this instrument, §3.3.)

Revision History

Versi on	Date	Author	Description
0.1	June 2026	T. Roxey	Initial version. DBA-MA layer. Three scenarios: Scenario A (cyber-kinetic turbo-expander attack), Scenario B (sour gas H ₂ S mass casualty), Scenario C (Permian Basin corridor cluster). Dual Design Basis architecture with four AI governance interaction points. Series-first: Bright Line must constrain commercial APC optimization logic as well as adversarial manipulation—five regulatory gap recommendations.
0.2	June 2026	T. Roxey	Full three-layer instrument. DB layer added (§3): non-adversarial bounding events across equipment failure, natural hazard, and operational event categories. DBA layer added (§4): non-adversarial consequence envelope, PPSD acceptance criterion framework established with Zone 1/2/3 escalation structure, Tier 1/2/3 consequence taxonomy applied. MA extension (§5) updated to reference DB/DBA foundation. Parent document reference updated to DBA-GAS Sector Framework v1.6 and DBA-EN-SF v0.6. Gap 2.2 in the DBA-EN series gap register closed.